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Wu

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(54) **OIL TANK CAP**

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(51) **Int. Cl.**

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B65D 41/02	(2006.01)
B65D 51/16	(2006.01)
B65D 53/02	(2006.01)

(52) **U.S. Cl.**

CPC ***F01M 11/04*** (2013.01); ***B65D 41/02*** (2013.01); ***B65D 51/1644*** (2013.01); ***B65D 53/02*** (2013.01); *B65D 2205/02* (2013.01); *B65D 2251/205* (2013.01); *F01M 2011/0491* (2013.01)

(58) **Field of Classification Search**

CPC F01M 11/04; F01M 2011/0491; B65D 41/02; B65D 51/1644; B65D 53/02; B65D 2205/02; B65D 2251/205

See application file for complete search history.

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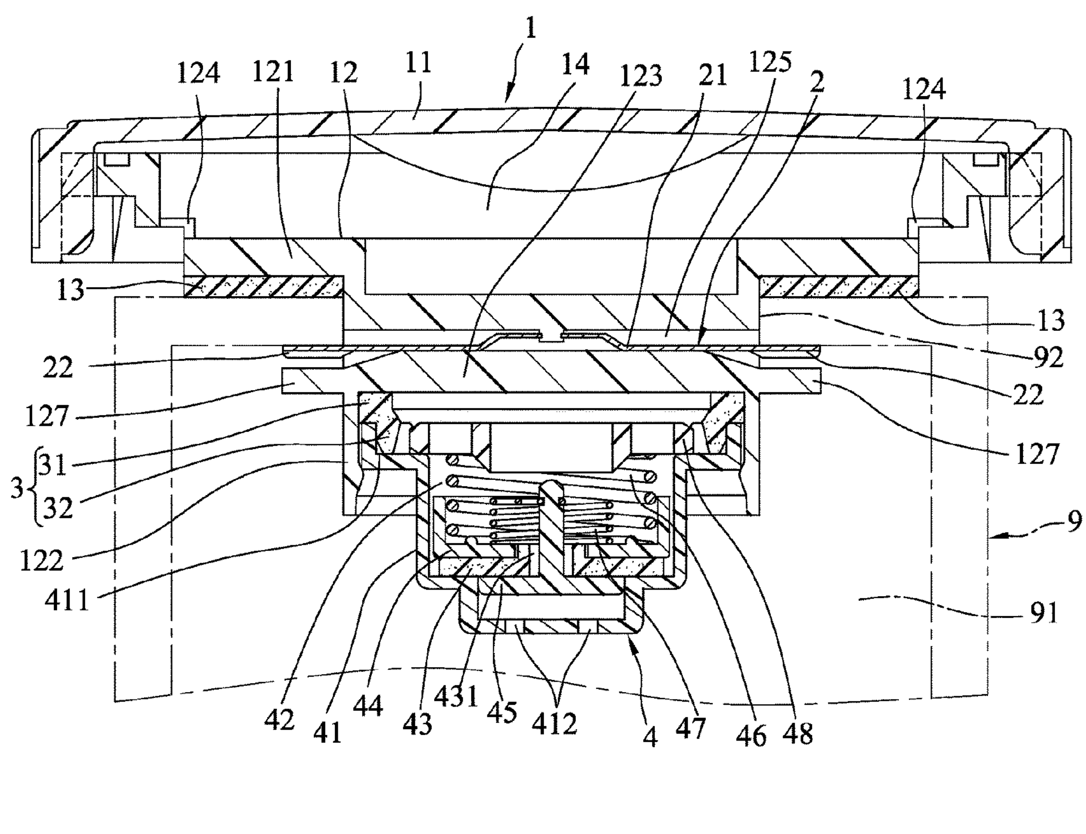
(57) **ABSTRACT**

An oil tank cap adapted to be removably engaged with an oil tank having an oil storing space. The oil tank cap includes a cap body, an air seal, and an air valve. The cap body includes an outer casing, and a cap base being plastic and one-piece, and defining an accommodation space with the outer casing. The cap base has an annular seat portion that has at least one exhaust hole, a tube portion, a separation portion having at least one ventilation hole. The air seal has a protrusion portion, and an annular body that encircles the ventilation holes. The air valve is inserted into the tube portion, is connected airtightly to the air seal, and is operable to open or close for balancing pressure of the oil storing space and pressure of an external environment.

8 Claims, 7 Drawing Sheets



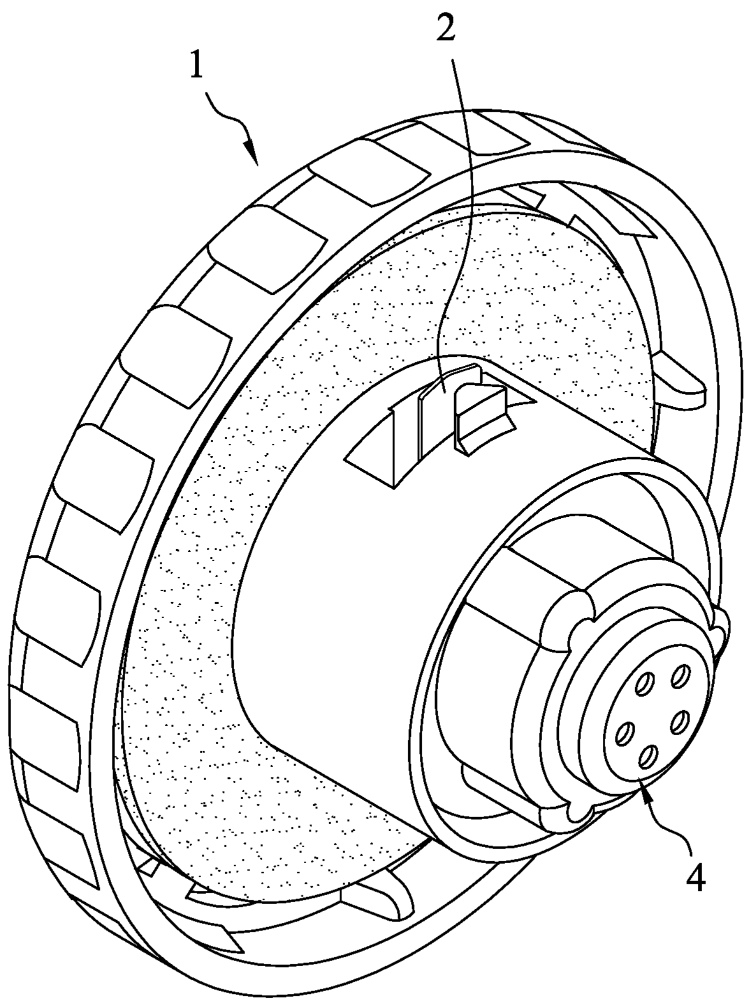


FIG. 1

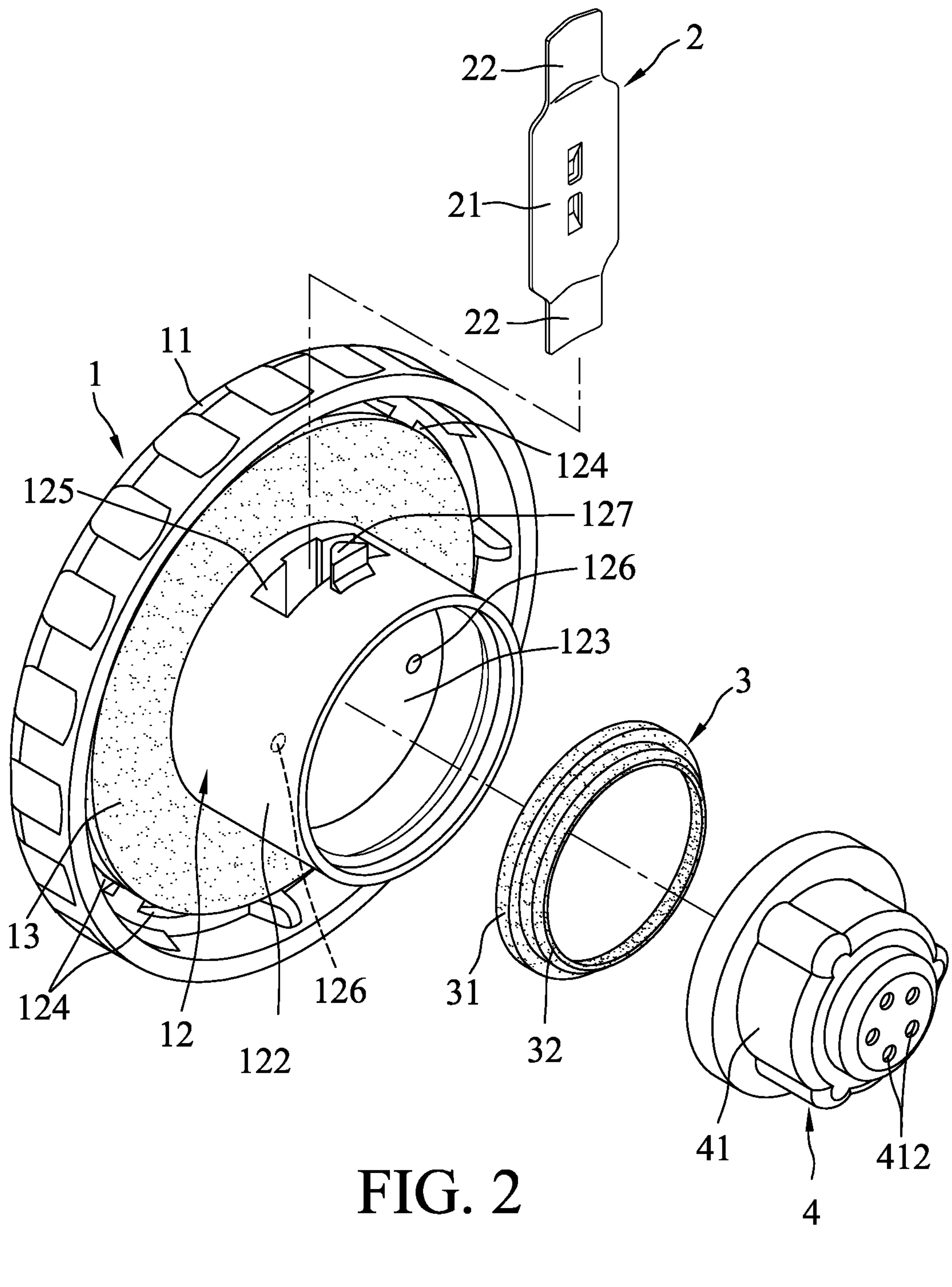


FIG. 2

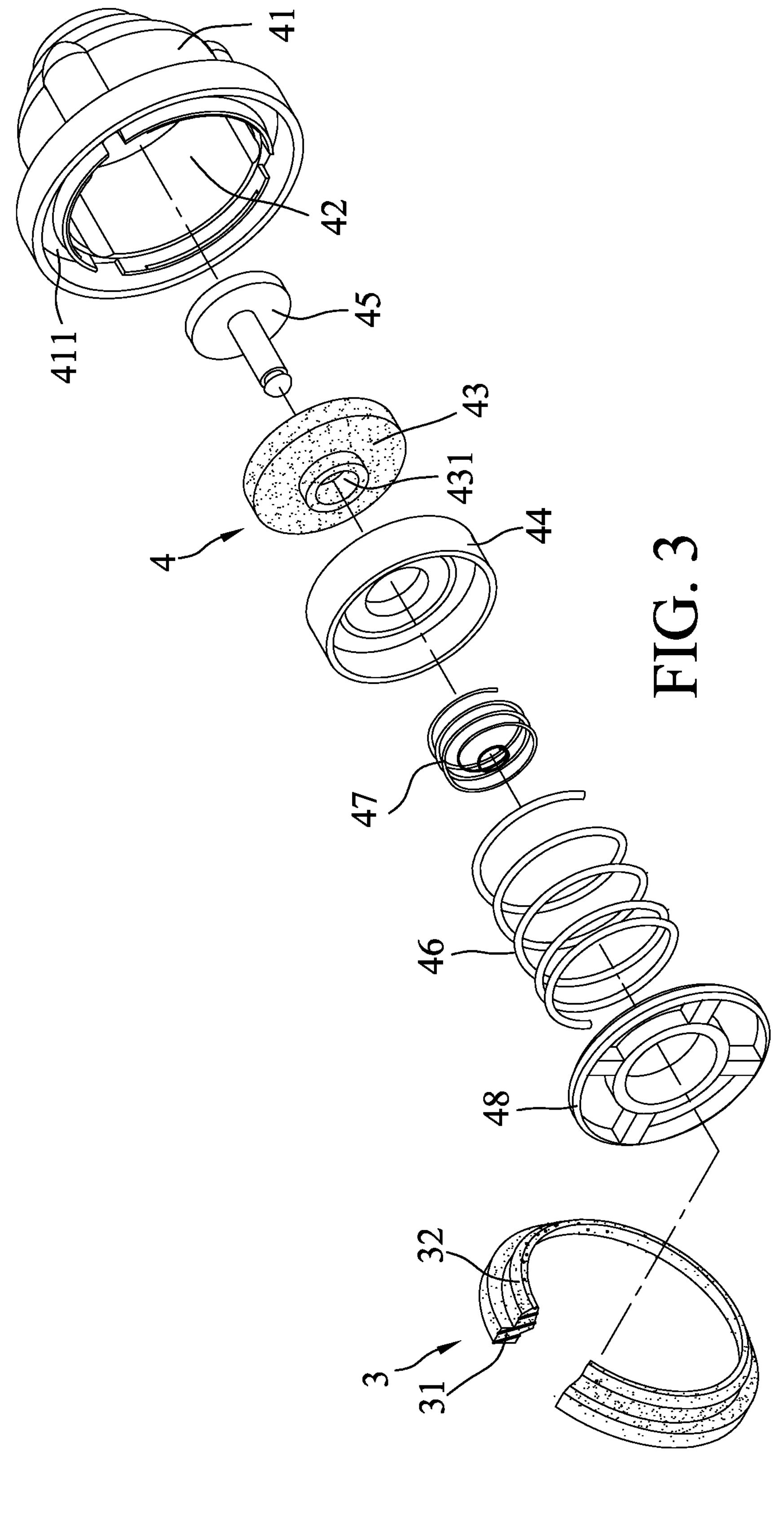


FIG. 3

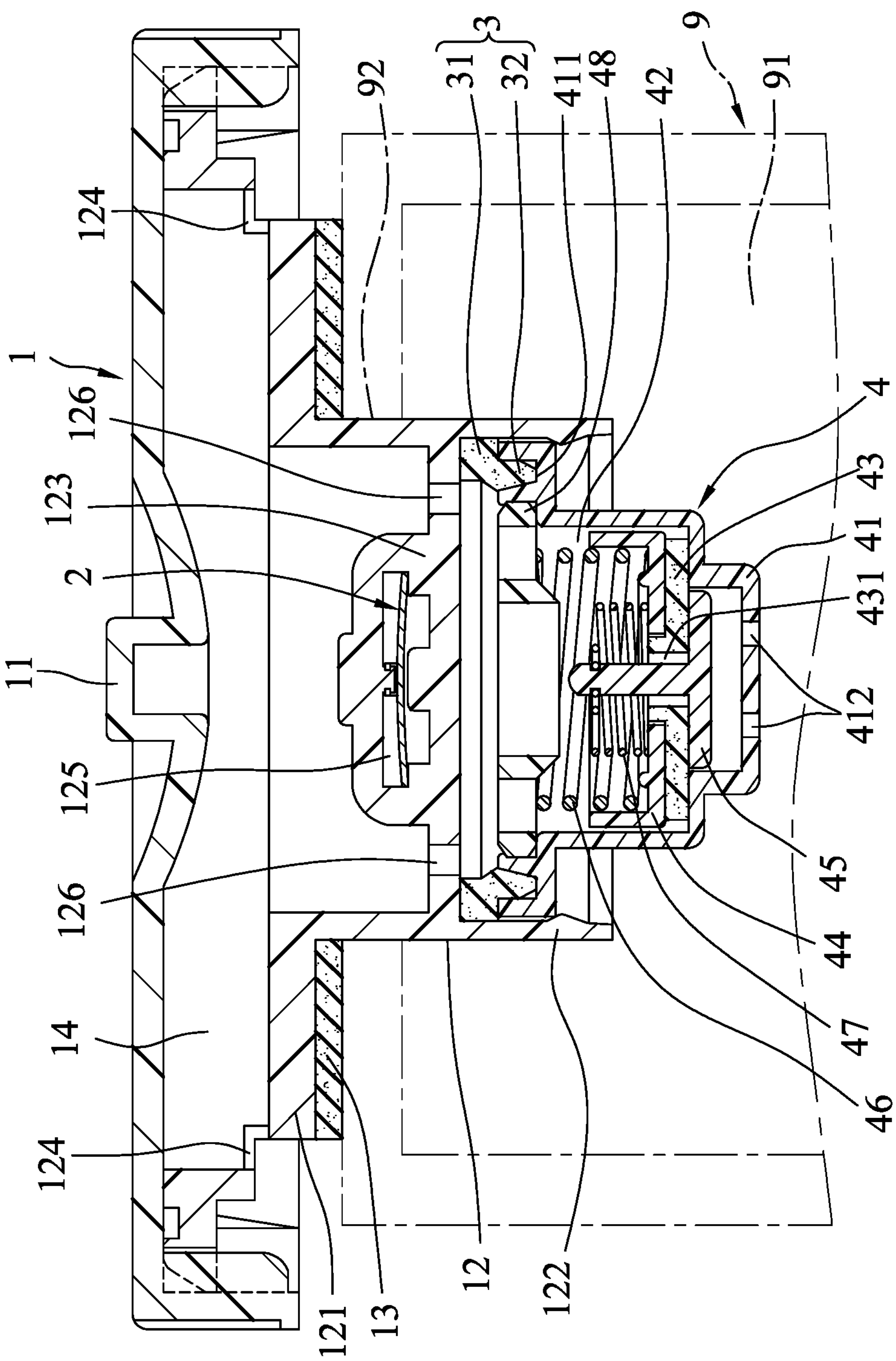

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FIG. 4

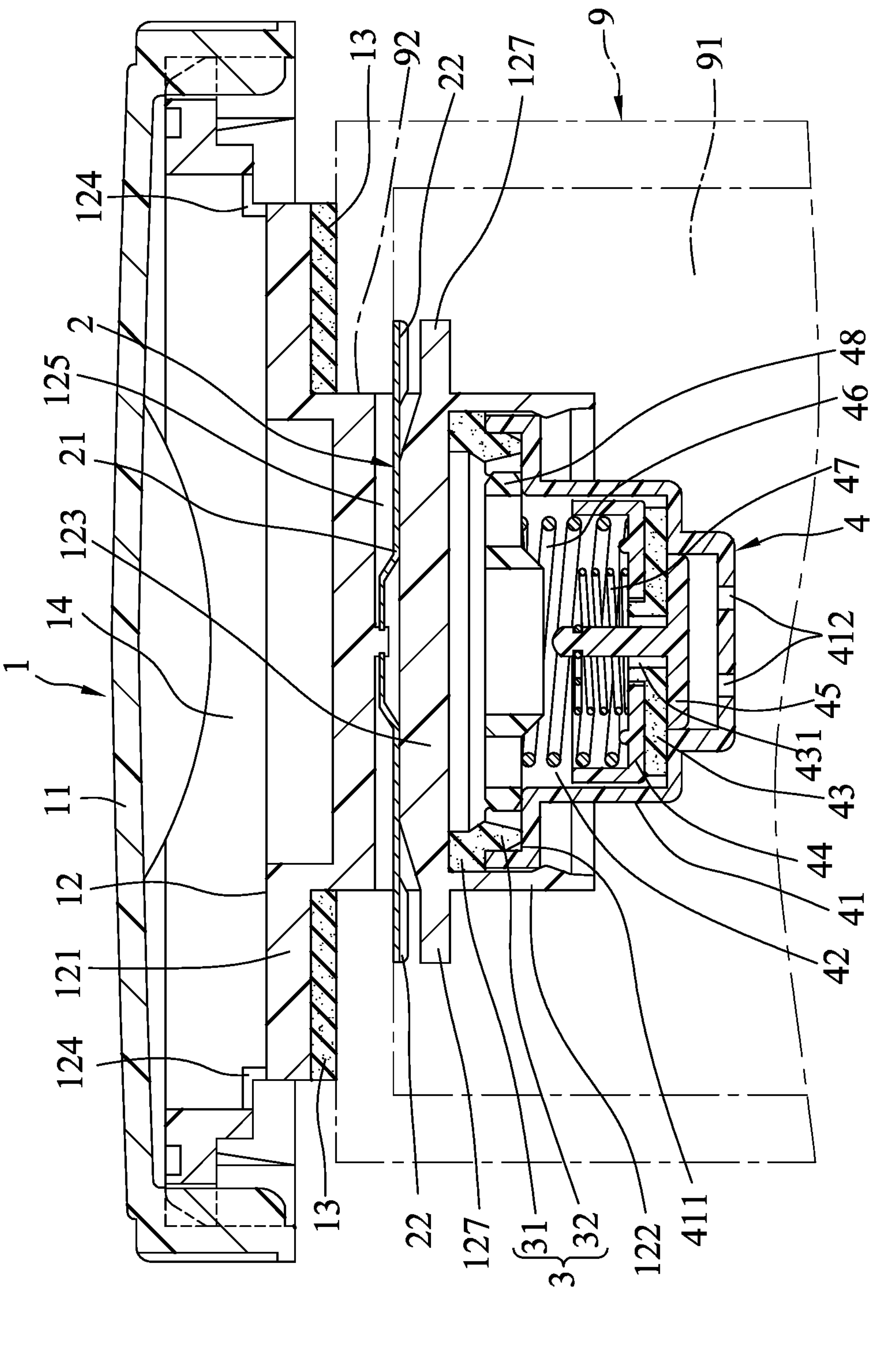
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FIG.5

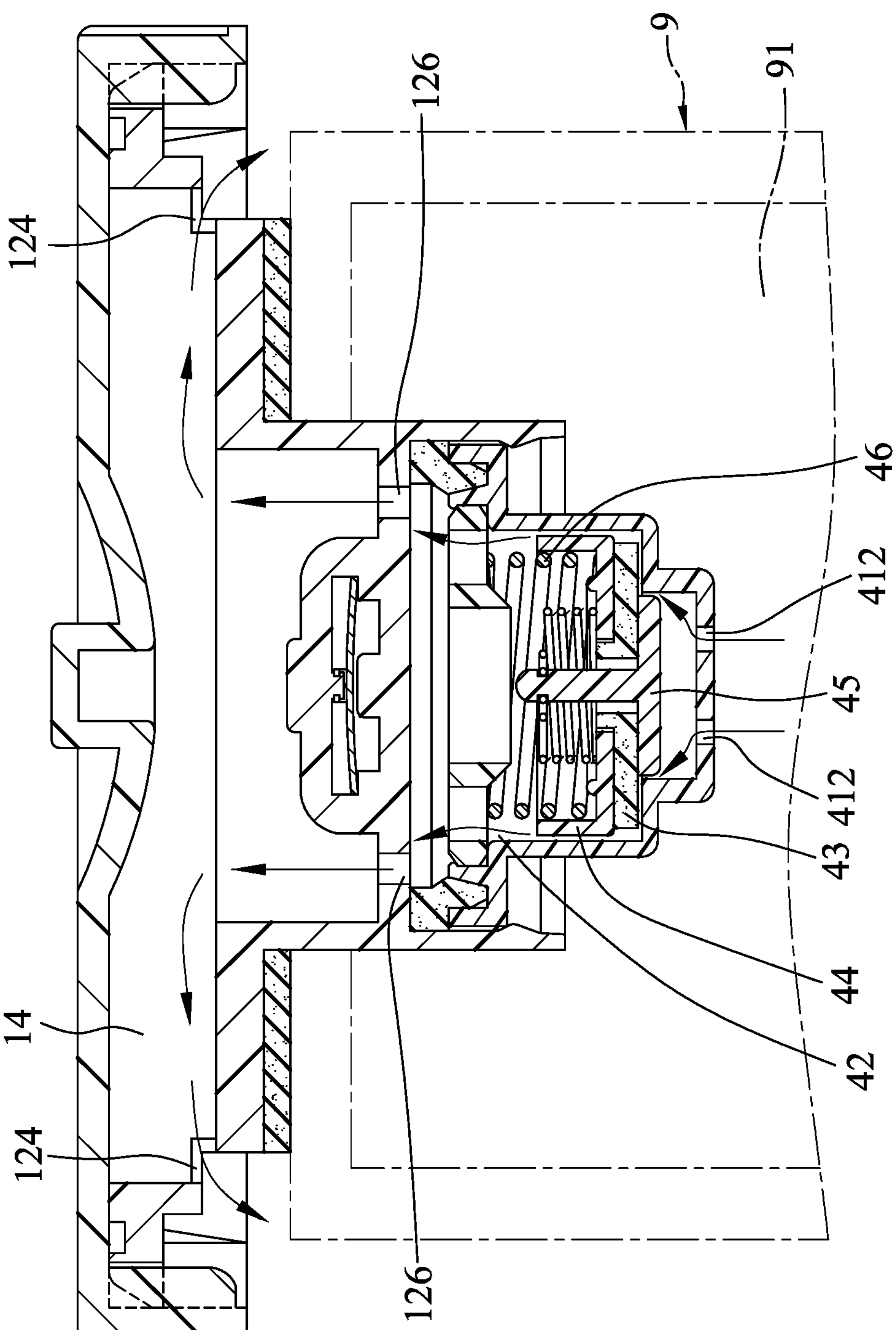

FIG. 6

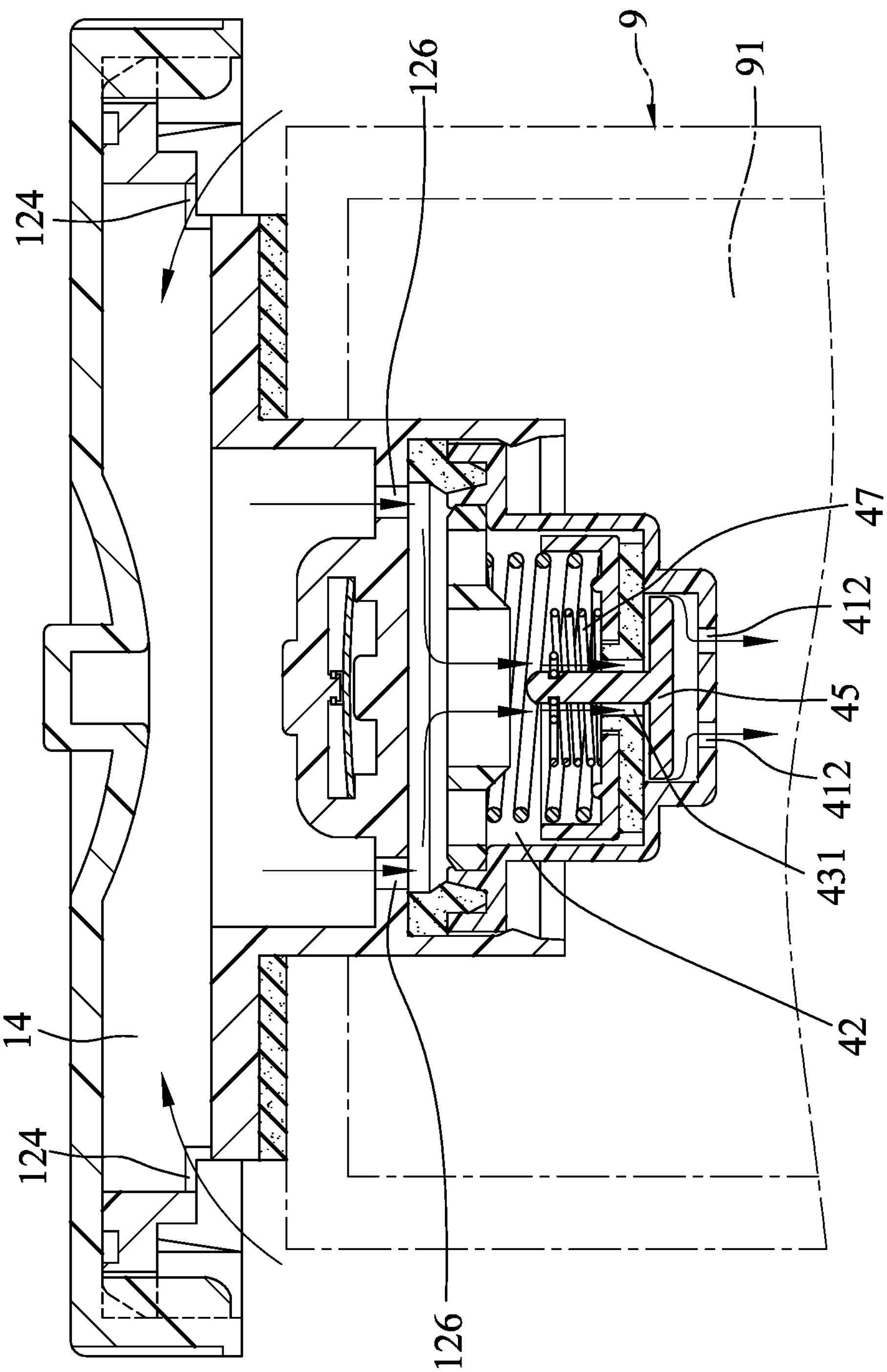

FIG. 7

OIL TANK CAP

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority to Taiwanese Utility Model Patent Application No. 111212217, filed on Nov. 8, 2022, the entire disclosure of which is incorporated by reference herein.

FIELD

The disclosure relates to an oil tank cap, and more particularly to an oil tank cap adapted to be removably engaged with an oil inlet of an oil tank.

BACKGROUND

A conventional oil tank cap is adapted to be removably engaged with an oil inlet of an oil tank, and includes a cap, a cap base that cooperatively defines an accommodating space with the cap, and other components. The manufacturing cost of the cap body of the conventional oil tank cap is significant due to the cap body being composed of a plurality of components that are riveted together by rivets. Furthermore, when the cap body is hit by a force, stress gathers around the riveted junctions, which may break the rivets or cause the components to be loose, thereby producing safety issues. Hence, there is room for improvement on the conventional oil tank cap.

SUMMARY

Therefore, an object of the disclosure is to provide an oil tank cap that can alleviate at least one of the drawbacks of the prior art.

According to the disclosure, the oil tank cap oil tank cap is adapted to be removably engaged with an oil tank. The oil tank has an oil storing space, and an oil inlet that spatially communicates with the oil storing space. The oil tank cap includes a cap body, a spring strip, an air seal, and an air valve. The cap body includes an outer casing and a cap base. The cap base is plastic and one-piece, and cooperates with the outer casing to define an air accommodation space. The cap base has an annular seat portion, a tube portion, and a separation portion. The annular seat portion is adapted to be disposed outside the oil inlet, and has an inner periphery, an outer periphery opposite to the inner periphery and connected to the outer casing, and at least one exhaust hole interconnecting the air accommodation space and an external environment. The tube portion extends from the inner periphery of the annular seat portion, and is adapted to extend into the oil inlet. The separation portion is connected to and disposed inside the tube portion. The air accommodation space is between the separation portion and the outer casing. The separation portion has an accommodating slot and at least one ventilation hole. The accommodating slot is formed in a middle of the separation portion, and is isolated from the air accommodation space. The at least one ventilation hole is disposed beside and spaced apart from the accommodating slot, and is adapted for interconnecting the air accommodation space and the oil storing space. The spring strip extends through the accommodating slot, and is adapted to be disposed in the oil storing space and adjacent to the oil inlet. The air seal is disposed in the tube portion of the cap body, abuts against the separation portion, and encircles the at least one ventilation hole. The air valve is inserted into the tube portion, is connected airtightly to the air seal, and is operable to open or close for balancing pressure of the oil storing space and pressure of the external environment.

BRIEF DESCRIPTION OF THE DRAWINGS

Other features and advantages of the disclosure will become apparent in the following detailed description of the embodiment(s) with reference to the accompanying drawings. It is noted that various features may not be drawn to scale.

FIG. **1** is a perspective view illustrating an embodiment of an oil tank cap according to the disclosure.

FIG. **2** is an exploded perspective view illustrating the embodiment.

FIG. **3** is a fragmentary exploded perspective view of an air seal and an air valve of the embodiment.

FIG. **4** is a sectional view of the embodiment, illustrating the air valve being closed.

FIG. **5** is another sectional view of the embodiment, illustrating the air valve being closed.

FIG. **6** is a sectional view similar to FIG. **4**, but illustrating the air valve being opened to release air.

FIG. **7** is another sectional view similar to FIG. **4**, but illustrating the air valve being opened to intake air.

DETAILED DESCRIPTION

Before the disclosure is described in greater detail, it should be noted that where considered appropriate, reference numerals or terminal portions of reference numerals have been repeated among the figures to indicate corresponding or analogous elements, which may optionally have similar characteristics.

It should be noted herein that for clarity of description, spatially relative terms such as “top,” “bottom,” “upper,” “lower,” “on,” “above,” “over,” “downwardly,” “upwardly” and the like may be used throughout the disclosure while making reference to the features as illustrated in the drawings. The features may be oriented differently (e.g., rotated 90 degrees or at other orientations) and the spatially relative terms used herein may be interpreted accordingly.

Referring to FIGS. **1**, **2**, and **4**, an embodiment of the oil tank cap is adapted to be removably engaged with an oil tank **9** of a vehicle, for example, an automobile (not shown). The oil tank **9** has an oil storing space **91** that is for storing fuel or lubricant, and an oil inlet **92** that spatially communicates with the oil storing space **91**. The oil tank cap includes a cap body **1**, a spring strip **2**, an air seal **3**, and an air valve **4**.

Referring to FIGS. **2**, **4**, and **5**, the cap body **1** includes an outer casing **11** for gripping, a cap base **12** cooperating with the outer casing **11** to define an air accommodation space **14**, and a washer **13** sleeved on the cap base **12**.

Specifically, the cap base **12** is made of a plastic material and is manufactured as one-piece. The manufacturing process may be implemented by, for example, molding. The cap base **12** has an annular seat portion **121**, a tube portion **122**, and a separation portion **123**. The annular seat portion **121** is adapted to be disposed outside the oil inlet **92**, and has an inner periphery and an outer periphery that is opposite to the inner periphery and that is connected to the outer casing **11**. The tube portion **122** extends from the inner periphery of the annular seat portion **121**, and is adapted to extend into the oil inlet **92**. The separation portion **123** is connected to and disposed inside the tube portion **122**. The air accommodation space **14** is between the separation portion **123** and the outer casing **11**. In this embodiment, the annular seat portion **121** has eight exhaust holes **124** that are disposed around an inner periphery of the outer casing **11**, that are equiangularly spaced apart, and that interconnect the air accommodation space **14** and an external environment. It should be noted that the number of the exhaust holes **124** is not limited to eight, and may vary as needed. The separation portion **123** has a flat surface at a bottom end thereof, and a protruding portion at a top end thereof with two recesses disposed on opposite sides of the protruding portion. The separation portion **123** further has an accommodating slot **125** that is formed in a middle thereof and that is isolated from the air accommodation space **14**, and two ventilation holes **126** that are disposed beside and spaced apart from the accommodating slot **125** and that are adapted for interconnecting the air accommodation space **14** and the oil storing space **91**. The washer **13** is sleeved on the tube portion **122**, and is adapted to be cushioned between the annular seat portion **121** and the oil tank **9**. The cap base **12** further has two limiting portions **127**. Each of the limiting portions **127** extends from the tube portion **122**, is disposed lower than a respective one of opposite ends of the accommodating slot **125**, and has a triangular cross section. It should be noted that the number of the ventilation holes **126** are not limited to two, and the number may vary as needed.

The spring strip **2** is made of a metal material, extends through the accommodating space **125**, and is adapted to be disposed in the oil storing space **91** and adjacent to the oil inlet **92**. The spring strip **2** includes a positioning portion **21** disposed in the accommodating slot **125**, and two abutting portions **22** extending respectively from opposite ends of the positioning portion **21** and extending outwardly of the accommodating slot **125**. Each of limiting portions **127** of the cap base **12** is aligned with a respective one of the abutting portions **22** of the spring strip **2**. Specifically, each of the abutting portions **22** is adapted to be disposed between the oil inlet **92** and a respective one of the limiting portions **127**, and is adapted to abut against the inner surface of the oil tank **9** when the cap body **1** is rotated to be locked to the oil tank **9**. Resilient deformation of the abutting portions **22** may be limited by the limiting portions **127**. Specifically, when the cap body **1** or a surrounding area of the oil inlet **92** is hit by a force, the abutting portions **22** deform and abut respectively against the limiting portions **127**, so that excessive deformation of the abutting portions **22** may be prevented. It should be noted that each of the abutting portions **22** of the spring strip **2** has an inverted V shape and complements each of the limiting portions **127**. In other embodiments, the abutting portions **22** may be shaped differently.

The air seal **3** is ring-shaped and disposed in the tube portion **122** of the cap body **1**, and includes an annular body **31** that abuts against the separation portion **123** and that encircles the ventilation holes **126**, and a protruding portion **32** that extends from the annular body **31** and away from the separation portion **123** of the cap base **12**.

The air valve **4** includes a valve base **41** inserted into the tube portion **122** and abutting tightly against the air seal **3**. The valve base **41** and the separation portion **123** cooperatively define a control space **42** that is spatially connected to the ventilation holes **126**. The valve base **41** has an annular groove **411** that is engaged airtightly with the protruding portion **32** of the air seal **3**, and five air inlet holes **412** that are formed on a bottom of the valve base **41** and that are adapted for interconnecting the control space **42** and the oil storing space **91**. It should be noted that the number of the air inlet holes **412** is not limited to five, and may vary as needed in other embodiments.

Referring to FIGS. **3** to **5**, the air valve **4** further includes a piston **43** movably disposed in the control space **42** and having a central hole **431**, a spring seat **44** disposed on the piston **43** between the piston **43** and the separation portion **123**, a pin rod **45** having a head portion and a pin portion, a large spring **46**, a small spring **47**, and a hollow plate **48**. Specifically, the head portion of the pin rod **45** separably abuts against the piston **43** and is aligned with the air inlet holes **412**, and the pin portion of the pin rod **45** extends co-movably from the head portion through the central hole **431** of the piston **43**. The large spring **46** is mounted on the spring seat **44** for biasing the spring seat **44** towards the air inlet holes **412**. The small spring **47** is connected between the spring seat **44** and the rod portion of the pin rod **45** for biasing the pin rod **45** away from the air inlet holes **412**. The hollow plate **48** is disposed inside the valve base **41**. The large spring **46** has opposite ends abutting respectively against the hollow plate **48** and the spring seat **44**.

The air valve **4** is operable to open or close for balancing pressure of the oil storing space **91** and pressure of the external environment.

Referring to FIG. **6**, the piston **43** is movable against a resilient force of the large spring **46** to permit or block passage of air, which enters the control space **42** via the inlet holes **412**, from the control space **42** to the ventilation holes **126**. For example, when a fuel or a lubricant volatilizes (i.e., passes off in vapor) inside the oil storing space **91**, thereby causing the pressure of the oil storing space **91** to be higher than that of the external environment, the vapor enters the control space **42** through the ventilation holes **412**, pushes against the head portion of the pin rod **45** and opens the air valve **4** by upwardly moving the pin rod **45**, the piston **43**, and the spring seat **44**, thereby creating a gap among the pin rod **45**, the piston **43** and the inner surface of the valve base **41**, and allowing the vapor to enter the control space **42** via the gap, to advance into the air accommodation space **14** via the ventilation holes **126**, and to finally flow into the external environment through the exhaust holes **124**. When the pressure of the oil storing space **91** and the pressure of the external environment are balanced, the pin rod **45** is released, and the large spring **46** resiliently biases the spring seat **44** towards the air inlet holes **412** to close the air valve **4**.

Referring to FIG. **7**, the pin rod **45** is movable against a resilient force of the small spring **47** to permit or block passage of air, which enters the control space **42** via the ventilation holes **126**, from the control space **42** to the air inlet holes **412** via the central hole **431** of the piston **43**. For example, when a large amount of the fuel or lubricant is extracted from the oil tank **9**, thereby causing the pressure of the oil storing space **91** to be lower than that of the external environment, the pin rod **45** is downwardly drawn towards the air inlet holes **412**, thereby creating a gap between the pin rod **45** and the piston **43**, and allowing an airflow from the external environment to enter the control space **42** via the exhaust holes **124**, to advance into the control space **42** via the ventilation holes **126**, and to finally flow through the central hole **431**, the gap, and the air inlet holes **412**. After the pressure of the oil storing space **91** and the pressure of the external environment are balanced, the pin rod **45** is released, and the small spring **47** resiliently biases the pin rod **45** to close the air valve **4**.

The advantages of the oil tank cap of the disclosure is as follows: By manufacturing the cap base **12** as one-piece, the structural strength and the safety of the oil tank cap increases, and the manufacturing cost is reduced. Hence, the objective of the disclosure is achieved.

Furthermore, by engaging the protruding portion **32** of the air seal **3** with the annular groove **411** of the valve base **41**, the valve base **41** is sealed airtightly to the air seal **3**, thereby preventing leaking of air or vapor between the valve base **41** and the tube portion **122**. The protrusion portion **32** further has the function of positioning the air seal **3** and preventing deformation of the air seal **3**, which may otherwise block the ventilation holes **126**.

In the description above, for the purposes of explanation, numerous specific details have been set forth in order to provide a thorough understanding of the embodiment(s). It will be apparent, however, to one skilled in the art, that one or more other embodiments may be practiced without some of these specific details. It should also be appreciated that reference throughout this specification to "one embodiment," "an embodiment," an embodiment with an indication of an ordinal number and so forth means that a particular feature, structure, or characteristic may be included in the practice of the disclosure. It should be further appreciated that in the description, various features are sometimes grouped together in a single embodiment, figure, or description thereof for the purpose of streamlining the disclosure and aiding in the understanding of various inventive aspects; such does not mean that every one of these features needs to be practiced with the presence of all the other features. In other words, in any described embodiment, when implementation of one or more features or specific details does not affect implementation of another one or more features or specific details, said one or more features may be singled out and practiced alone without said another one or more features or specific details. It should be further noted that one or more features or specific details from one embodiment may be practiced together with one or more features or specific details from another embodiment, where appropriate, in the practice of the disclosure.

While the disclosure has been described in connection with what is(are) considered the exemplary embodiment(s), it is understood that this disclosure is not limited to the disclosed embodiment(s) but is intended to cover various arrangements included within the spirit and scope of the broadest interpretation so as to encompass all such modifications and equivalent arrangements.

What is claimed is:

1. An oil tank cap adapted to be removably engaged with an oil tank, the oil tank having an oil storing space, and an oil inlet that spatially communicates with the oil storing space, said oil tank cap comprising:
- a cap body that includes
  - an outer casing, and
  - a cap base being plastic and one-piece, cooperating with said outer casing to define an air accommodation space, and having
    - an annular seat portion that is adapted to be disposed outside the oil inlet, and that has an inner periphery, an outer periphery opposite to said inner periphery and connected to said outer casing, and at least one exhaust hole interconnecting said air accommodation space and an external environment,
    - a tube portion that extends from said inner periphery of said annular seat portion, and that is adapted to extend into the oil inlet, and
    - a separation portion that is connected to and disposed inside said tube portion, said air accommodation space being between said separation portion and said outer casing, said separation portion having
      - an accommodating slot that is formed in a middle of said separation portion, and that is isolated from said air accommodation space, and
      - at least one ventilation hole that is disposed beside and spaced apart from said accommodating slot, and that is adapted for interconnecting said air accommodation space and the oil storing space;
- a spring strip that extends through said accommodating slot, and that is adapted to be disposed in the oil storing space and adjacent to the oil inlet;
- an air seal that is disposed in said tube portion of said cap body, that abuts against said separation portion, and that encircles said at least one ventilation hole; and
- an air valve that is inserted into said tube portion, that is connected airtightly to said air seal, and that is operable to open or close for balancing pressure of the oil storing space and pressure of the external environment.

2. The oil tank cap that is claimed in claim **1**, wherein:
- said air seal includes
  - an annular body abutting against said separation portion and encircling said at least one ventilation hole, and
  - a protruding portion extending from said annular body and away from said separation portion of said cap base; and
- said air valve includes a valve base inserted into said tube portion of said cap base, abutting tightly against said air seal, and having an annular groove that is engaged airtightly with said protruding portion of said air seal.

3. The oil tank cap as claimed in claim **2**, wherein:
- said valve base and said separation portion cooperatively define a control space that is spatially connected to said at least one ventilation hole; and
- said valve base further has at least one air inlet hole that is adapted for interconnecting said control space and the oil storing space.

4. The oil tank cap as claimed in claim **3**, wherein:
- said air valve further includes
  - a piston movably disposed in said control space, and having a central hole,
  - a spring seat disposed on said piston between said piston and said separation portion,
  - a pin rod having a head portion that separably abuts against said piston and that is aligned with said at least one air inlet hole, and a pin portion that extends co-movably from said head portion through said central hole of said piston,
  - a large spring mounted on said spring seat for biasing said spring seat towards said at least one air inlet hole, and
  - a small spring connected between said spring seat and said rod portion of said pin rod for biasing said pin rod away from said at least one air inlet hole;
- said piston is movable against a resilient force of said large spring to permit or block passage of air, which enters said control space via said at least one air inlet hole, from said control space to said at least one ventilation hole; and
- said pin rod is movable against a resilient force of said small spring to permit or block passage of air, which enters said control space via said at least one ventilation hole, from said control space to said at least one air inlet hole via said central hole of said piston.

5. The oil tank cap as claimed in claim **1**, wherein:
- said spring strip includes

a positioning portion disposed in said accommodating slot, and

two abutting portions extending respectively from opposite ends of said positioning portion and extending outwardly of said accommodating slot.

6. The oil tank cap as claimed in claim **5**, wherein:

said cap base further has two limiting portions, each of said limiting portions extending from said tube portion, being disposed lower than a respective one of opposite ends of said accommodating slot, and being aligned with a respective one of said abutting portions of said spring strip for limiting resilient deformation of said abutting portions.

7. The oil tank cap as claimed in claim **6**, wherein:

each of said limiting portions of said cap base has a triangular cross section; and

each of said abutting portions of said spring strip has an inverted V shape.

8. The oil tank cap as claimed in claim **1**, wherein said cap body further includes a washer that is sleeved on said tube portion.

* * * * *